

L8: LFR Benchmark

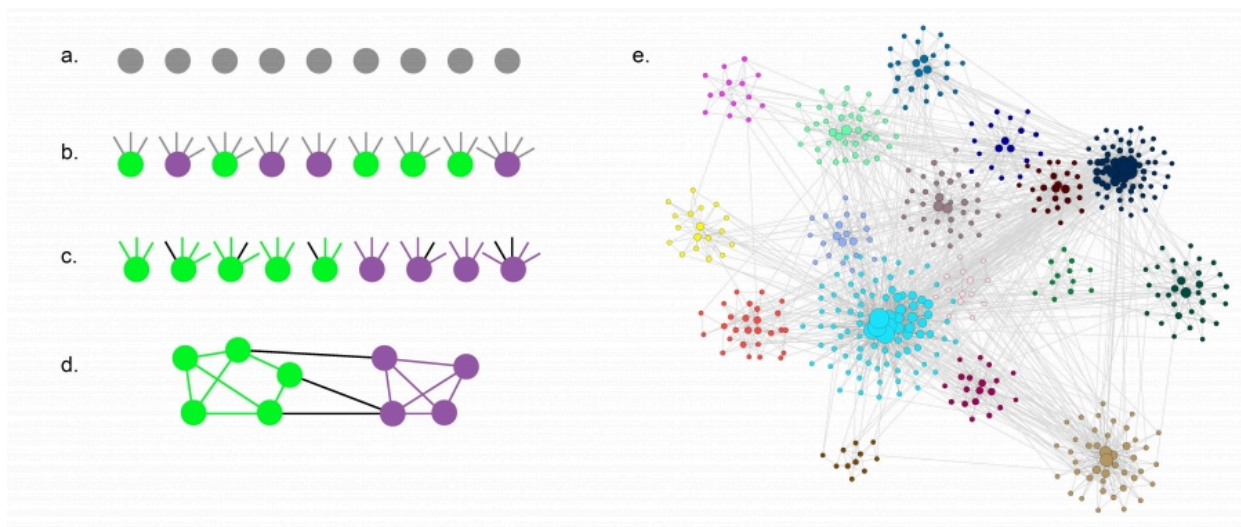


Figure 9.26 from *Network Science* by Albert-László Barabási

The LFR benchmark (*Lancichinetti-Fortunato-Radicchi*) is more realistic because it generates networks with scale-free degree distribution – and additionally, the community size follows a power-law size distribution.

We need to specify the total number of nodes n , the parameter μ that we also saw in the GN benchmark, two exponent parameters: γ for the node degree distribution and ζ for the community size distribution, as well as the minimum and maximum values for the degree distribution and the community size distribution.

LFR starts with n disconnected nodes, giving each of them a degree (*number of stubs*) based on the degree distribution.

Also, each node is assigned to a community based on the community size distribution (*see part-a and part-b*).

The stubs of each node are then classified as either internal, to be connected to other internal stubs of nodes in the same community – or external, to be connected to available external stubs of nodes in other communities (*see part-c*). This stub partitioning process is controlled by the parameter μ of the GN benchmark.

The connections between nodes are then assigned randomly as long as a node has available stubs of the appropriate type (*internal vs external*) (*see part-d*).

The visualization part-e shows an LFR network with $n=500$, $\mu=0.4$, $\gamma=2.5$, and $\zeta=2$. Note that the 16 communities have very different sizes, and the nodes are also highly heterogeneous in terms of degree.